

Task 17 — Customer Segmentation (RFM) Report

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1. EXECUTIVE SUMMARY & OBJECTIVE

Objective: Transform an undifferentiated base of 150 active customers into 5 actionable, strategic segments using Recency, Frequency, and Monetary (RFM) modeling.

Methodology: Scores were computed via quartile ranking (qcut/rank) on heavily right-skewed transaction logs to prevent equal-width binning distortion.

2. SEGMENT PROFILES & RECOMMENDED ACTIONS

SEGMENT NAME	SIZE (N)	AVG RECENCY	AVG FREQ	AVG MONETARY	STRATEGIC RECOMMENDATION
Champions	42	24.1 Days	9.8 Tx	,842.50	VIP Loyalty & Referral Perks
Loyal Customers	35	38.5 Days	7.2 Tx	,120.30	Upsell & Cross-sell Programs
Recent / Promising	28	19.2 Days	2.1 Tx	.40	Welcome Series & Product Tours
At-Risk / Need Attention	25	112.4 Days	6.8 Tx	.10	Win-Back Discounts & Surveys
Hibernating / Churned	20	165.8 Days	1.4 Tx	.20	Low-Cost Automated Re-engage

3. PITFALL SAFEGUARDS & STABILITY VALIDATION

- **Quantile Binning:** Used quartile rank transformations rather than fixed-width bins to accommodate highly right-skewed customer spending profiles.
- **Action Alignment:** Every segment has an explicit commercial play tied to customer lifetime value.
- **Maintenance Plan:** Automated a 30-day scheduled refresh pipeline to prevent segment decay.

4. DELIVERABLES & REPOSITORY VERIFICATION

- **Output Datasets:** rfm_transactions.csv and rfm_segmentation_results.csv.
- **GitHub Repository:** Live on <https://github.com/JeevanN-07/placemux.git>.